

QiNoteOS_Dev_Doc_v1.2_ColdStart_Final

Below is the **QiNoteOS Developer Documentation v1.2 (Cold-Start Complete)**.

This version is **loss-proof**: if all chat history vanished, a competent engineer could rebuild the system correctly from scratch without guessing intent.

Save as:

```
docs/QiNoteOS_Dev_Doc_v1.2_ColdStart_Final.md
```

QiNoteOS — Developer Documentation (v1.2, Cold-Start Complete)

System: QiNoteOS

Status: Phase 1  Complete · Phase 2  Locked & Operational

Date: 2026-01-01

Authority: This document supersedes all prior Phase 1–2 dev notes.

0. Purpose of This Document

This document must satisfy **three guarantees**:

1. A new engineer can start from **zero** (blank machine) and reach a working system.
2. Architectural boundaries cannot be accidentally violated.
3. Future contributors cannot “optimize” the system into contradiction.

If something is ambiguous, it is a defect in this document.

1. Non-Negotiable Philosophy (Locked)

QiNoteOS is a **filesystem-first knowledge operating system**.

Five axioms:

1. The **filesystem** is the only source of truth

2. Identity (QID) > path names
3. Derived artifacts are disposable
4. Orchestration ≠ authorship
5. AI is assistive, never authoritative

Violation of any axiom invalidates the system.

2. High-Level Architecture (Authoritative)

```
QiNoteOSv1/  
├─ app/           # Tools, kernel, UI (replaceable)  
├─ vault/         # Canonical knowledge (portable, truth)  
├─ docs/          # Governance & dev documentation  
└─ versions/      # Global changelog
```

Layer Responsibilities

Layer	Role	May Do	Must Not Do
Vault	Truth	Store content	Generate artifacts
qid-cli	Identity	Issue/validate IDs	Orchestrate
Kernel	Orchestration	Validate, index, serve	Author content
Lite Viewer	Presentation	Render modules	Mutate truth
AI (future)	Assistance	Suggest, retrieve	Decide truth

3. Cold-Start Bootstrap (From Zero)

3.1 System Requirements

- OS: Windows, macOS, or Linux
 - Python: **3.11+** (tested with 3.11–3.14)
 - No Node, no database, no Docker required
-

3.2 Clone & Prepare

```
git clone <repo-url> QiNoteOSv1
cd QiNoteOSv1
```

Create virtual environment:

```
python -m venv .venv
```

Activate:

Windows

```
.venv\Scripts\Activate.ps1
```

macOS / Linux

```
source .venv/bin/activate
```

3.3 Install Tooling

```
pip install -r app/tools/qid-cli/requirements.txt
```

3.4 Directory Sanity Check (MANDATORY)

You must have:

```
QiNoteOSv1/
├─ app/
├─ vault/
├─ docs/
```

If `vault/` does not exist, create it manually:

```
mkdir vault
mkdir vault/modules
```

! The app must never “own” the vault.

4. QID System (Phase 1 — Sealed)

4.1 What qid-cli Is

- **Identity authority**
- Stateless except for its registry
- Portable

4.2 What qid-cli Is NOT

- Not an orchestrator
- Not a UI
- Not a daemon
- Not called automatically by kernel

4.3 Canonical Commands

Run from:

```
app/tools/qid-cli/
```

Initialize registry:

```
python qid.py init
```

Create a root ID:

```
python qid.py new-root --band PRODUCT --title "How to Use QiNoteOS"
```

Create a module skeleton:

```
python qid.py module-init \  
  --root qidXXXXXXX_0 \  
  --slug how_to_use_qinoteos \  
  --out ../../../../vault/modules \  
  --title "How to Use QiNoteOS"
```

Validate a module:

```
python qid.py validate ../../../../vault/modules/qidXXXXXXX_how_to_use_qinoteos
```

5. Module Contract (v1 — Locked)

5.1 Required Files

```
module.yaml  
README.md  
content.md  
assets/
```

5.2 Optional / Generated

```
versions/  
snapshots/  
links/  
viewer/
```

5.3 Rules

- Assets must use **relative paths**
 - Module must be copy-paste portable
 - No absolute references
 - No external dependencies
-

6. Kernel (Phase 2 — Locked)

6.1 Kernel Authority

The kernel is a **single orchestrator process**.

There are **no per-module daemons**.

6.2 Kernel Invocation (CRITICAL)

Always run as a module:

```
python -m app.kernel.src.kernel <command>
```

Never run `kernel.py` directly.

6.3 Kernel Commands

Build derived indexes:

```
python -m app.kernel.src.kernel build-index --vault ./vault --clean
```

Watch for changes:

```
python -m app.kernel.src.kernel watch --vault ./vault --build-on-start
```

Serve dev + viewer access:

```
python -m app.kernel.src.kernel serve --vault ./vault --host 0.0.0.0 --port 8787
```

6.4 Kernel Output

```
vault/_derived/  
├─ vault_index.json  
└─ modules/  
    └─ <module>/  
        └─ module_index.json
```

`_derived/` may be deleted at any time.

7. Lite Viewer (Module-Local, Offline)

7.1 Purpose

- Human-readable rendering
- Offline capable (`file://`)

- No kernel required to view
- No frameworks

7.2 Location (Locked)

```
vault/modules/<module>/viewer/
```

7.3 Viewer Files

```
index.html
styles.css
app.js
data.js
build_data.py
serve.py
```

7.4 Viewer Lifecycle Contract (CRITICAL)

Action	Who	Allowed
Scaffold viewer	Kernel	✓ if missing only
Overwrite viewer	Kernel	✗
Populate data.js	User / script	✓
Auto-regenerate data	Kernel	✗
Modify canonical files	Viewer	✗

Viewer population is **explicit and intentional**:

```
python viewer/build_data.py
```

8. Command Authority Matrix (Sealed)

Operation	qid-cli	Kernel	Viewer	User
Issue QID	✓	✗	✗	✗
Init module	✓	✗	✗	✗
Validate module	✓	✓	✗	✗
Build indexes	✗	✓	✗	✗
Watch filesystem	✗	✓	✗	✗
Scaffold viewer	✗	✓	✗	✗
Populate viewer data	✗	✗	✗	✓
Edit content	✗	✗	✗	✓

If a tool exceeds its column, it is a bug.

9. Versioning & Snapshots

Module-Local Log

```
versions/log.md
```

- Human-authored
- Explains intent

Snapshots

```
snapshots/
```

- Known-good states
 - Auditable
 - Never edited
-

10. Git Strategy (Required)

Root `.gitignore`

```
# Python
__pycache__/  
*.pyc  
.venv/  
  
# Derived artifacts  
vault/_derived/  
  
# Module runtime data  
vault/modules/*/logs/  
vault/modules/*/snapshots/  
  
# OS  
.DS_Store  
Thumbs.db
```

Kernel logs may be git-tracked if desired.

11. Known Failure Modes & Fixes (Tribal Knowledge Made Explicit)

Symptom	Cause	Fix
Relative import error	Ran kernel.py directly	Use <code>python -m</code>
Vault created under app	Path resolved incorrectly	Vault must be repo-root
Module marked invalid	Missing optional dirs	Create empty dirs
Viewer 404	UI path misresolved	Kernel resolves via app root
Lock file hangs	Stale lock	Delete <code>.lock</code> file

12. Roadmap (Aligned & Locked)

Phase	Focus
Phase 0	Identity & structure
Phase 1	Reader-first modules
Phase 2	Kernel orchestration

Phase	Focus
Phase 3	Embeddings (derived)
Phase 4	Orchestration scale
Phase 5	Workflow polish

No phase may violate earlier contracts.

13. Final Invariant (Write This Everywhere)

The filesystem is truth.
IDs are identity.
Orchestration never authors.
AI never decides.

14. Status

- Cold-start capable: ✓
- Loss-of-chat safe: ✓
- Refactor-resistant: ✓
- Phase 1–2 complete: ●

This document is now the **canonical engineering reference** for QiNoteOS v1.

If you want next, the only *correct* moves are:

- Freeze this as **v1.2 locked**
- Generate a **Decision Log (immutable)**
- Begin **Phase 3 contracts** (interfaces only)

Anything else risks architectural drift.